



3500 Engine Aftercooler Assembly Test

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The operation of the after-cooler system is critical for proper engine operation, and if there are any errors in the assembly of the system, it can lead to future leaks. At Finning Chile (CRC Antofagasta), we have created a test to detect any assembly errors to avoid any future redo and machine downtime in the field.

To seal the after-cooler group, we have fabricated tooling to permit the simulation of air pressure in the after-cooler group, and to check for any potential leakage.

2.0 Best Practice Description

- 2.1 Install the fabricated covers to the lower manifold and upper housing (some models) will require a hose connection at the upper housing utilizing a fabricated blocking plug in the hose.



- 2.2 Set the primary and secondary pressure gauges to 30 PSI.

Main gauge air line



Secondary gauge air line



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2.3 After setting the pressure gauges to 30 PSI, connect the air supply hose to the front of the lower manifold and open the valve and allow the air pressure to stabilize in the after-cooler group to 30 PSI.

Power line



Gauge pressure connected to the supply line 30 PSI



2.4 Once the air pressure has stabilized at 30 PSI, turn the valve off and wait for five minutes for any pressure decrease from 30 PSI and if no pressure decrease is observed after five minutes the assembled after-cooler group is good.



2.5 Release the air pressure and remove the tooling.

3.0 Implementation Steps

Items required for performing the aftercooler test:



1.- Upper hoses and plugs.
2.- Pressure gauge.

3.- Hose.
4.- Lower covers.

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4.0 Benefits

- Detect potential leaks quickly.
- Ensures quality assembly.
- Avoid unnecessary field repairs and machine downtime.

5.0 Resources Required

- Fabricate the elements.
- Staff training.

6.0 Supporting Attachments / References

- None so far.

7.0 Related Best Practices

- None so far.

8.0 Acknowledgements

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